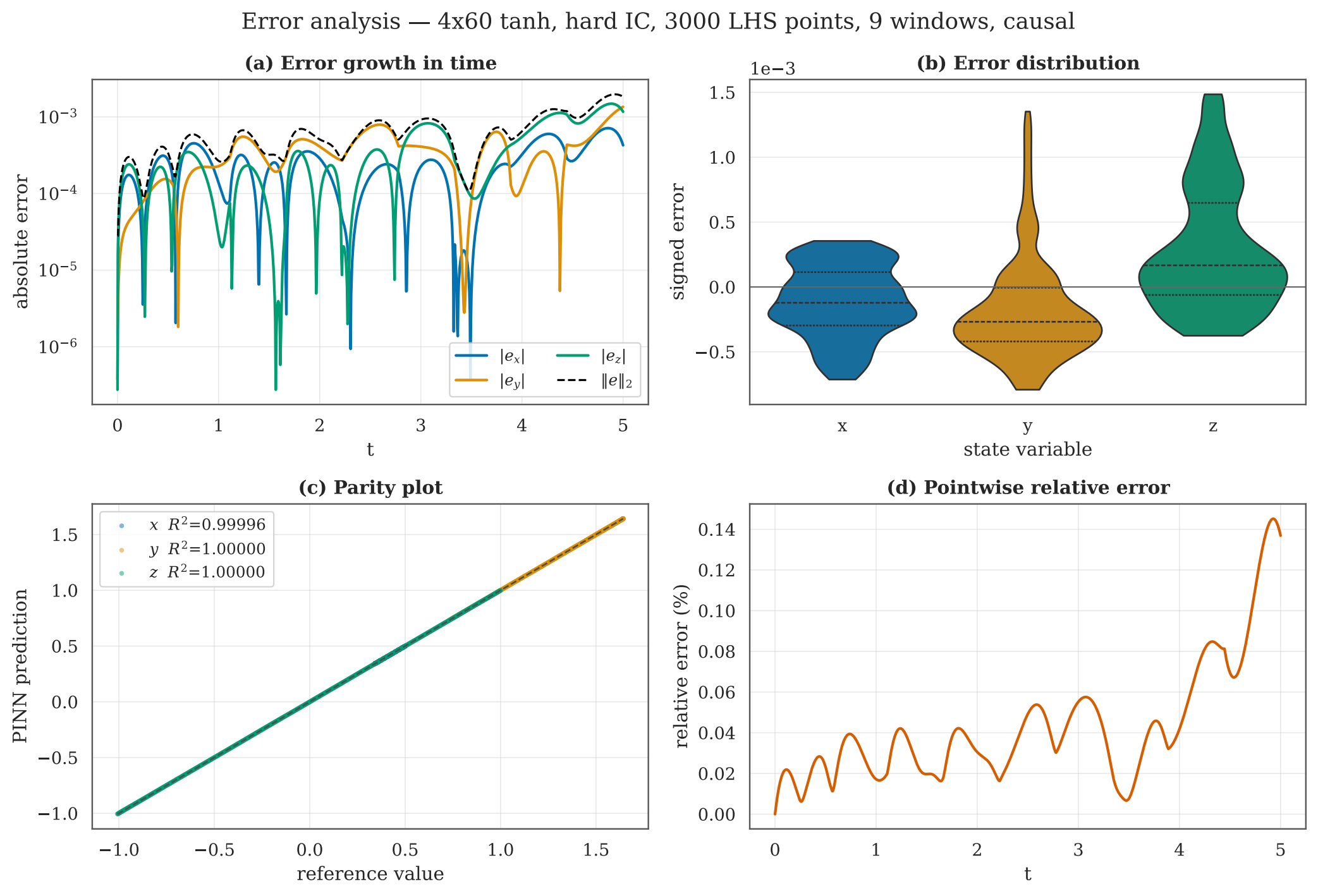

Error analysis — 4x60 tanh, hard IC, 3000 LHS points, 9 windows, causal
(a) Error growth in time
absolute error
t
$|e_x|$
$|e_y|$
$|e_z|$
$\|e\|_2$
(b) Error distribution
1e−3
signed error
x
y
z
state variable
(c) Parity plot
x R^2=0.99996
y R^2=1.00000
z R^2=1.00000
PINN prediction
reference value
(d) Pointwise relative error
relative error (%)
t